

Learning Metamaterial Eigenmodes with Wavelet-Encoded Fourier Neural Operators

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Abstract

Machine learning surrogates based on neural operators have shown broad applicability in solving forward PDE problems. However, eigenvalue problems, in which an eigenparameter and one of several valid eigenmodes must be simultaneously solved, remain difficult because standard operator learning formulations assume a unique input–output map. This work demonstrates that Fourier Neural Operators (FNOs), combined with wavelet-based encodings of PDE inputs, can learn and inexpensively predict multiple eigenmodes of the elastic wave equation, corresponding to deformation modes of acoustic waves propagating through arbitrary metamaterial geometries. We provide theoretical motivation and experimental evidence for why wavelet encodings are well matched to the dual spatial–spectral structure of the FNO, enabling deterministic mode selection on both continuous and discretely valued geometries within a single model, and for why prediction accuracy varies with geometric discontinuities. These results carry broader implications for designing input encodings in other multi-mode PDE solvers based on spectral neural operators.

Keywords: Neural Operator, Metamaterials, Wavelets, Eigenmodes, Machine Learning

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1. Introduction

The numerical solution of partial differential equations (PDEs) underpins modern scientific computing and engineering design. Applications ranging from structural mechanics and fluid dynamics to wave propagation and materials engineering routinely rely on computationally intensive numerical methods such as finite element analysis (FEA) to obtain accurate solutions. While these methods are highly reliable, their computational cost can become prohibitive in settings that require repeated evaluations, such as optimization, uncertainty quantification, inverse design, and real-time control [1, 2].

Recent advances in machine learning have introduced neural operators as a promising alternative to traditional numerical solvers. Unlike conventional neural networks that learn mappings between finite-dimensional vectors, neural operators learn mappings between function spaces, allowing them to approximate entire solution operators for families of PDEs [3, 4]. Models such as the Fourier Neural Operator (FNO) have demonstrated remarkable accuracy and computational efficiency across a variety of linear and nonlinear PDE problems while maintaining strong generalization capabilities across discretizations and geometries [5]. Related physics-informed operator architectures likewise seek rapid emulation of parametric PDE solution maps [6]. These developments have generated significant interest in using neural operators as surrogate models for computational physics applications.

Despite this progress, existing neural operator architectures have primarily been developed and evaluated on PDEs whose governing parameters are known a priori and whose solutions are uniquely determined by the provided inputs. A large class of physically important problems, however, are governed by eigenvalue PDEs, in which both the solution field and one or more unknown PDE parameters must be determined simultaneously. Examples include vibration analysis, quantum mechanics, photonics, acoustics, and wave propagation in metamaterials [7, 2]. In these problems, multiple valid solutions may exist for the same physical domain, corresponding to different eigenvalue-eigenvector pairs. The ability to represent and distinguish among these multiple solutions is essential for accurately modeling the underlying physics.

Acoustic metamaterials provide a particularly compelling example of this challenge. These architected materials derive their properties from geometric structure rather than composition alone, enabling engineered interactions with propagating waves [8, 9]. Their dispersive behavior is compactly described by phononic band structures arising from Bloch eigenvalue analysis of periodic unit cells [10, 2]. Understanding and designing such systems often requires computing multiple deformation modes associated with different resonant frequencies and wave behaviors. Traditionally, these modes are obtained through computationally expensive eigenvalue analysis using finite element methods; even characterizing dispersion under material and geometric variability can require large ensembles of repeated FEA solves [1]. A neural operator capable of accurately predicting multiple deformation modes directly from material geometry would significantly accelerate such iterative design and analysis processes.

However, extending neural operators to eigenvalue PDEs is nontrivial. Standard operator-learning formulations implicitly assume a one-to-one mapping between inputs and outputs, whereas eigenvalue problems admit multiple valid solutions for a given geometry. As a result, existing FNO architectures lack a mechanism for deterministically selecting among different eigenmodes. Furthermore, the representation of eigenvalue-related information must be compatible with the spectral architecture of the FNO, which processes information jointly in both spatial and frequency domains.

Prior machine-learning work on phononic and acoustic metamaterials has largely targeted dispersion curves, bandgaps, or scalar acoustic responses rather than full Bloch displacement eigenfields with deterministic multi-mode selection. Early neural networks predicted one-dimensional phononic dispersion from a few material parameters [11], and convolutional networks have been applied to phononic eigenvalue problems [7]. Surrogate models based on Gaussian process regression can accelerate dispersion-relation evaluation for architected materials [12], while neural operators have been used to predict sonic-crystal transmission-loss spectra [13] and wavevector-conditioned phononic band structures [14]. Broader reviews summarize deep-learning approaches for phononic-crystal and metamaterial design [15, 16, 17], and complementary interpretable machine-learning methods extract unit-cell design rules for targeted bandgaps [18, 19]. The present work differs by learning a single

107 FNO that maps geometry and wavelet-encoded modal parameters to both complex
108 displacement eigenmodes and eigenfrequencies across continuous and discrete unit
109 cells.

110 In this work, we introduce a framework that enables Fourier Neural Operators to
111 solve eigenvalue PDEs associated with elastic wave propagation in acoustic metama-
112 terials. The key idea is to augment the model inputs with wavelet-based encodings
113 of PDE parameters that uniquely identify individual eigenmodes while remaining
114 compatible with the FNO architecture. We demonstrate that these encodings allow
115 a single neural operator to learn multiple valid solutions corresponding to distinct
116 deformation modes and to reproduce those modes deterministically on demand. We
117 further show that conventional spatial encodings and purely spectral encodings do
118 not provide the same capability, highlighting the importance of the proposed wavelet
119 representation. We also evaluate the same model on continuous- and discrete-valued
120 geometries, recovering both displacement fields and eigenfrequencies, and examine
121 how geometric discontinuities interact with the truncated spectral structure of the
122 FNO.

123 The contributions of this paper are threefold:

- 124 1. We demonstrate that a single Fourier Neural Operator can learn multiple
125 eigenmodes across continuous and discrete metamaterial geometries, recovering
126 both eigenvectors (displacement fields) and eigenvalues (eigenfrequencies).
- 127 2. We introduce wavelet encodings of modal parameters and explain theoretically
128 why their spatial-spectral structure enables deterministic mode selection in
129 FNOs.
- 130 3. We show that prediction accuracy degrades with geometric discontinuities and
131 interpret this through spectral truncation, with implications for applying FNOs
132 to problems with sharp interfaces.

133 Together, these results extend the applicability of neural operators beyond con-
134 ventional forward PDE problems and establish a pathway toward efficient surrogate

135 modeling of eigenvalue-driven physical systems, including acoustic metamaterials and
136 other wave-based engineering applications.

137 2. Methodology

138 This section describes the methods used to generate data and train Fourier Neural
139 Operators for acoustic metamaterial eigenmode prediction. We first summarize the
140 FNO architecture, then define the metamaterial design space and the Bloch-FEA
141 simulations used for supervised labels. We next introduce the wavelet encodings of
142 modal parameters, describe dataset construction, and detail the training procedure.

143 2.1. Fourier Neural Operator

144 The principal theoretical motivation for Fourier Neural Operators is their universal
145 approximation property over function spaces. Let $\mathcal{A}$ and $\mathcal{U}$ denote Banach spaces of
146 input and output functions, respectively, and let

$$\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U} \tag{1}$$

147 be a continuous nonlinear operator. It has been shown that, given sufficient latent
148 width and Fourier modes, an FNO can approximate $\mathcal{G}$ arbitrarily well on compact
149 subsets of $\mathcal{A}$ [3]. Unlike conventional neural networks, which approximate finite-
150 dimensional functions, FNOs approximate operators between infinite-dimensional
151 function spaces by learning a sequence of global integral operators parameterized in
152 the Fourier domain [5].

153 This theoretical framework assumes that both the input and output are continuous
154 fields of infinite resolution. In practice, however, these fields are sampled on a finite
155 computational grid. The continuous Fourier transform is therefore replaced by a
156 discrete Fourier transform (DFT), and only a finite number of Fourier modes are
157 retained during training [5, 20]. Consequently, the learned model approximates the
158 continuous solution operator through its discrete representation while preserving the
159 same spectral operator-learning architecture.

160 For the present work, the operator acts on discretized fields over a 32×32 pixel
 161 spatial grid. The input field consists of three channels containing encoded information
 162 for the metamaterial geometry, the stimulatory wavevectors, and the band number.
 163 The output field consists of five channels corresponding to the predicted eigenvalue, or
 164 frequency, and eigenvector, or displacement field, of the resulting deformation mode.
 165 Thus, the discretized learned operator takes the form

$$\mathcal{G}_\theta : \mathbb{R}^{32 \times 32 \times 3} \rightarrow \mathbb{R}^{32 \times 32 \times 5}. \quad (2)$$

166 The discretized FNO follows the standard lift-transform-project architecture:

- 167 1. **Lifting:** The input field is embedded into a higher-dimensional latent space
 168 through a pointwise lifting operator,

$$v_1(x) = \ell(a(x)), \quad \ell : \mathbb{R}^3 \rightarrow \mathbb{R}^{d_L}. \quad (3)$$

- 169 2. **Fourier layers:** A sequence of Fourier layers is applied to the latent represen-
 170 tation. Each layer updates the hidden field according to

$$v_{j+1}(x) = \sigma \left(\mathcal{F}^{-1} [R_j(k) \mathcal{F}[v_j](k)] + W_j v_j(x) + b_j \right), \quad (4)$$

171 where $\mathcal{F}$ denotes the discrete Fourier transform, $R_j(k)$ is the learnable spectral
 172 kernel, W_j is a learnable pointwise linear transformation, b_j is a learnable bias
 173 term, and σ is the activation function. GELU was used in this work because it
 174 is smooth and differentiable.

- 175 3. **Projection:** The final latent representation is projected back to the desired
 176 output dimension through a pointwise projection operator,

$$u(x) = p(v_n(x)), \quad p : \mathbb{R}^{d_L} \rightarrow \mathbb{R}^5. \quad (5)$$

177 FNOs are well suited for acoustic metamaterial simulation because the underlying
 178 physics is governed by spatially distributed wave equations, where the response

179 at one location depends on the global geometry of the unit cell and the imposed
 180 wavevector. The Fourier layers are designed to use spectral convolutions to efficiently
 181 capture long-range interactions and periodic spatial structure, while the pointwise
 182 transformations preserve local geometric information. Although FNOs are commonly
 183 used as surrogate models for fixed PDE solution operators, the same operator-learning
 184 framework can be extended to eigenvalue problems by conditioning the input on
 185 modal parameters, such as the wavevector and band index, and training the network
 186 to output both the associated eigenvalue and eigenvector field. In this formulation,
 187 the FNO learns a parameterized eigensolution operator mapping geometry and modal
 188 encodings to the frequency and deformation mode of the acoustic unit cell.

189 This process is illustrated graphically in Figure 1. For our experiments, the
 190 NeuralOp Python package was used, which provides FNO model definitions with
 191 customizable numbers of layers, hidden channels, and input and output tensor sizes.
 192 The hyperparameter combinations explored, as well as the optimized final FNO
 193 configuration, are shown in Table S1 in Section 2.6. These hyperparameters are used
 194 alongside the fixed parameters of the problem space: model height 32, model width
 195 32, input channels 3, and output channels 5.

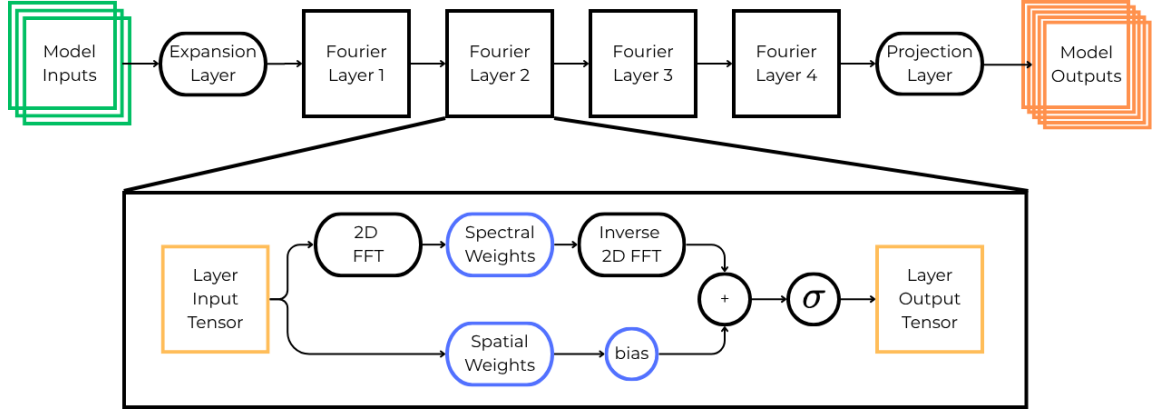


Figure 1: The FNO model architecture used in this project. The input data is first expanded according to the hidden-channels parameter, allowing multiple latent features to be represented between Fourier layers. Each Fourier layer is shown in detail in the expanded view. The data passing through a Fourier layer is processed through two learned branches: a pointwise branch, which applies a learned linear transformation in the spatial representation, and a spectral branch, which applies a fast Fourier transform (FFT), multiplies the retained Fourier modes by learned spectral weights, and then applies an inverse FFT. The outputs of both branches, along with a learnable bias term, are summed before passing through a nonlinear activation function. After passing through four Fourier layers, the latent representation is projected from the hidden-channel dimensionality to the final output dimensions.

2.2. Metamaterial Design Space

To train a surrogate model capable of predicting acoustic eigenmodes, a representative design space of acoustic metamaterials must first be defined. The design space was selected to provide sufficient geometric and material complexity to demonstrate operator learning for eigenvalue PDEs while maintaining computationally tractable finite element simulations for dataset generation.

The dataset is restricted to two-dimensional acoustic metamaterials composed of infinitely repeating square unit cells exhibiting eight-fold symmetry. Each unit cell is discretized onto a 32×32 spatial grid and represented by three material-property channels corresponding to the normalized elastic modulus, density, and Poisson’s ratio. Material properties are normalized to the interval $[0, 1]$ to improve numerical conditioning during training, while the corresponding physical values are recovered using the fixed material constants listed in Table 1. The two constituent materials were chosen to represent a stiff steel alloy and a compliant polymer.

Parameter	E_{polymer}	E_{steel}	ρ_{polymer}	ρ_{steel}	ν_{polymer}	ν_{steel}
Units	Pa	Pa	kg/m ³	kg/m ³	—	—
Value	1×10^6	2×10^9	1200	8000	0.45	0.3

Table 1: Material parameters. The hard material is chosen to be a common steel alloy and the soft material a rubbery polymer.

Unit-cell geometries are synthesized by sampling a Gaussian process with a periodic covariance kernel and subsequently enforcing eight-fold symmetry. This procedure produces smoothly varying spatial material distributions while ensuring compatibility with periodically tiled square lattices. The sampled field assumes values between 0 and 1, where 0 represents pure polymer, 1 represents pure steel, and intermediate values denote an effective mixture of the two materials below the spatial resolution of the discretization.

To investigate the influence of geometry representation on model performance, two classes of unit cells were generated. The first consists of the continuous-valued Gaussian process realizations described above and is referred to as the *continuous geometry* dataset. The second is obtained by thresholding the Gaussian process samples to binary values, producing geometries composed exclusively of polymer and steel. This second collection is referred to as the *discrete geometry* dataset. Each representation comprises approximately one-half of the total dataset.

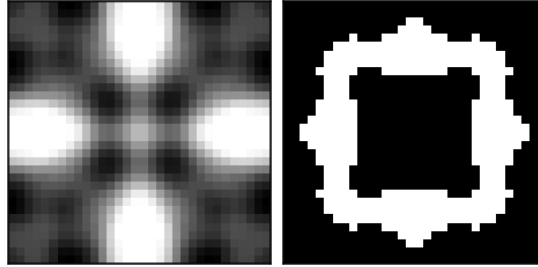


Figure 2: Example of a continuous geometry (left) and a discrete geometry (right) generated using the above process and parameters.

224 2.3. Wave Propagation Simulations

225 For each metamaterial geometry, wave propagation is simulated by solving the
 226 frequency-domain linear elastodynamic equation with finite element analysis (FEA),
 227 following the standard Bloch–Floquet treatment of periodic phononic media [2, 10].
 228 After spatial discretization and enforcement of Bloch periodicity, the governing
 229 problem takes the generalized eigenvalue form

$$(\mathbf{K}_r(\mathbf{k}) - \omega^2 \mathbf{M}_r(\mathbf{k})) \mathbf{u} = \mathbf{0}, \quad (6)$$

230 where ω is the angular eigenfrequency, $\mathbf{u}$ is the corresponding complex Bloch displace-
 231 ment eigenvector, and $\mathbf{k}$ is the wavevector in the irreducible Brillouin zone (IBZ). The
 232 reduced stiffness and mass matrices $\mathbf{K}_r(\mathbf{k})$ and $\mathbf{M}_r(\mathbf{k})$ are obtained from geometry-
 233 dependent global finite-element matrices $\mathbf{K}$ and $\mathbf{M}$ via a wavevector-dependent Bloch
 234 transformation matrix $\mathbf{T}(\mathbf{k})$,

$$\mathbf{K}_r(\mathbf{k}) = \mathbf{T}(\mathbf{k})^\dagger \mathbf{K} \mathbf{T}(\mathbf{k}), \quad \mathbf{M}_r(\mathbf{k}) = \mathbf{T}(\mathbf{k})^\dagger \mathbf{M} \mathbf{T}(\mathbf{k}), \quad (7)$$

235 with $(\cdot)^\dagger$ the Hermitian transpose. Here $\mathbf{T}(\mathbf{k})$ maps the full mesh degrees of freedom
 236 to an independent set while applying phase factors $e^{i\mathbf{k} \cdot \mathbf{R}}$ across periodic faces of the
 237 unit cell; $\mathbf{K}$ and $\mathbf{M}$ themselves depend only on the material field and are assembled
 238 once per geometry. These intermediate matrices are described briefly so the reader
 239 can compare the explicit FEA pipeline (Figure 3) with the Fourier neural operator
 240 architecture in Section 2.1; they are not inputs to the neural operator.

241 The IBZ is discretized into 25 x -direction and 13 y -direction wave numbers,
 242 totaling $25 \times 13 = 325$ unique wavevectors covering the IBZ half-plane. For each
 243 wavevector, the six lowest eigenfrequencies (bands) were computed using a custom
 244 FEA MATLAB script (see Section 4 for implementation details).

245 Each eigenvector $\mathbf{u}$ consists of two complex-valued displacement fields, (u_x, u_y) ,
 246 which are separated into their real and imaginary components to produce four
 247 real-valued output channels, $(u_{x,real}, u_{x,imag}, u_{y,real}, u_{y,imag})$. The corresponding eigen-
 248 frequency ω is encoded as a spatially uniform fifth channel, resulting in a target

249 tensor of size $5 \times 32 \times 32$. Combined with the three-channel material representation
 250 described in the previous subsection, each simulation therefore produces a supervised
 251 learning sample mapping

$$\mathbb{R}^{3 \times 32 \times 32} \rightarrow \mathbb{R}^{5 \times 32 \times 32}.$$

252 The overall finite element data-generation procedure is summarized in Figure 3.

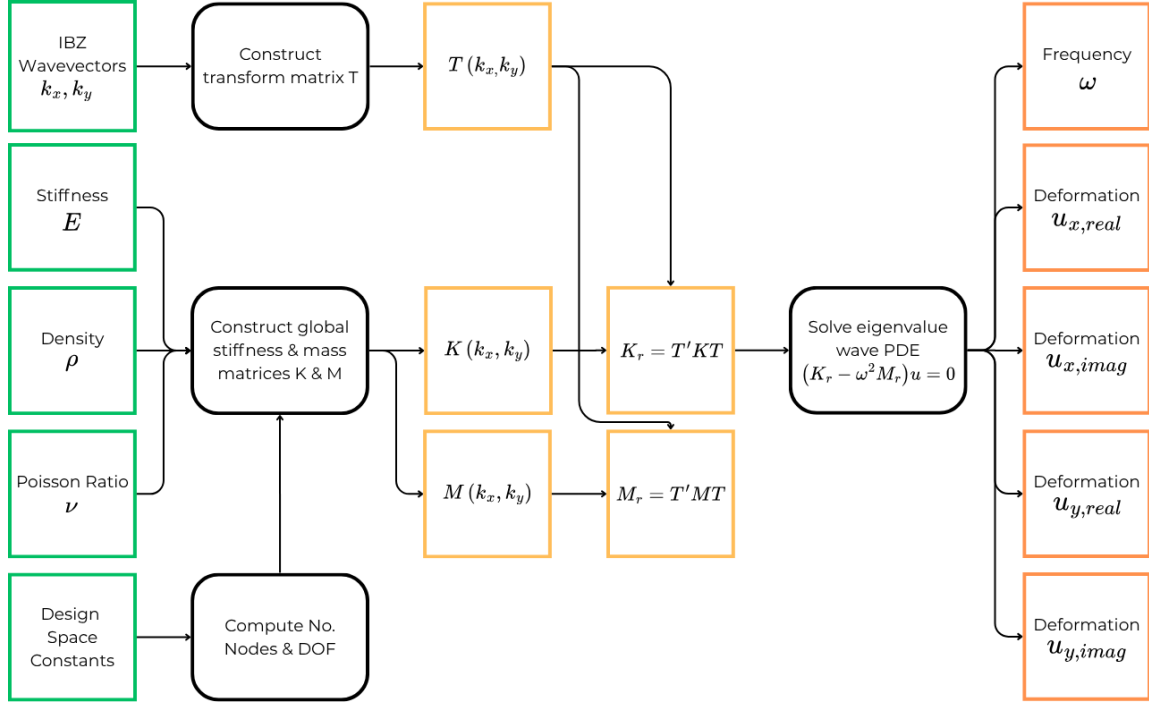


Figure 3: Flowchart of the FEA data generation method. Green squares represent varying inputs to the process, and orange squares represent computed quantities of the FEA process. Black rounded rectangles represent subroutines of the FEA process. The process starts with inputs of defined densities, stiffnesses, and Poisson ratios for each pixel of a metamaterial unit cell geometry (each a 32×32 pixel grid of values ranging from 0 to 1), along with a chosen k_x and k_y wavevector representing the stimulus to the metamaterial. A custom optimized FEA solver takes these inputs and produces intermediate matrices ($\mathbf{K}, \mathbf{M}, \mathbf{T}$), forms the reduced operators ($\mathbf{K}_r, \mathbf{M}_r$), and then solves Eq. (6) for the displacement fields (eigenvectors) and frequencies ω (eigenvalues) for the given inputs.

253 2.4. Input Wavelet Encoding

254 To allow the FNO model to toggle between PDE solutions for a given geom-
 255 etry, information must be fed to the model indicating which excitation waveform

256 (wavevectors) and deformation mode (band) we are looking for a solution to. This
 257 information is given in the form of a wavevector embedding, which we have found to
 258 yield superior results compared with a constant field embedding (2D matrices of the
 259 same value) or a sinusoidal embedding (2D sinusoidal fields with constants encoded
 260 as the sinusoidal frequency). A discussion on why wavelet encodings are superior is
 261 found in Section 3.4.

262 For the wavelet encodings, we use a 1D Gabor wavelet transform to map band
 263 numbers to unique wavelet images (see Algorithm 1), and a 2D Gabor wavelet
 264 transform to map x and y wavevectors to another set of unique wavelet images (see
 265 Algorithm 2), following the classical construction of jointly localized space–frequency
 266 atoms [21]. These representations show rich variation both in the spatial domain
 267 and in the Fourier domain, which allows for FNO model weights of both branches to
 268 richly interact with the wavelet representation, leading to better learning outcomes.

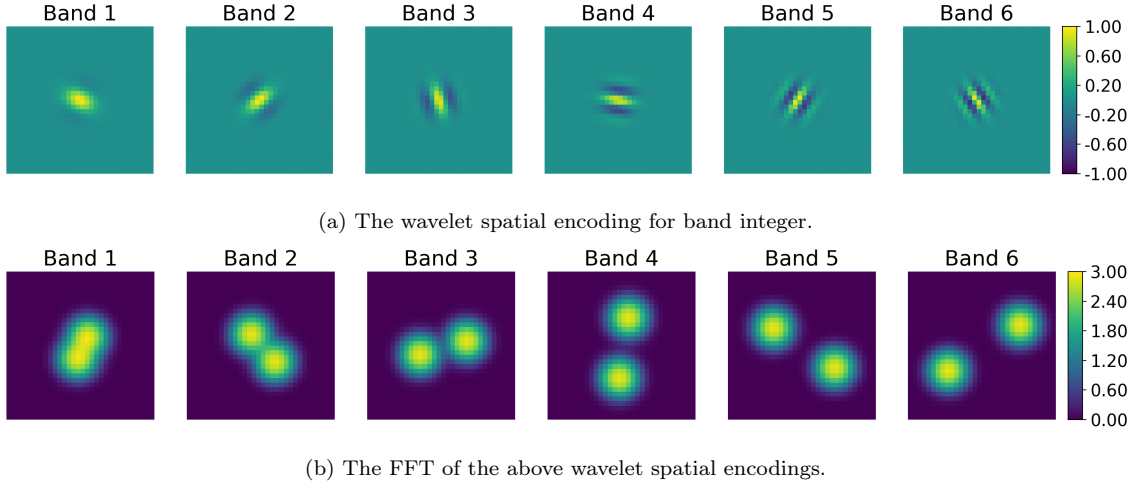
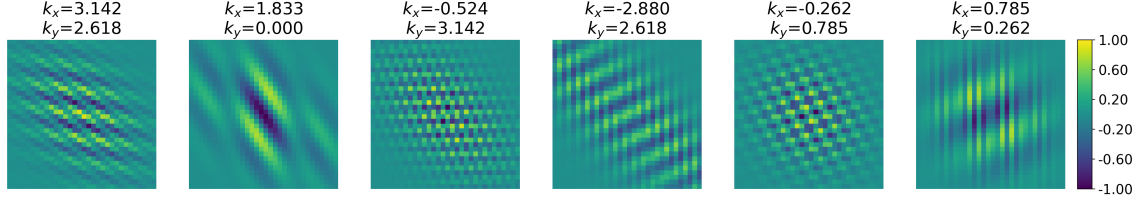
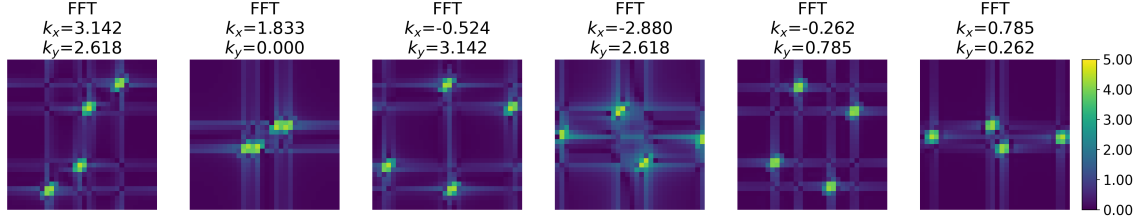


Figure 4: The wavelet encoding of the band integer (Fig. 4a) and its FFT (Fig. 4b). Each band corresponds to a unique wavelet with representation in both spatial and spectral domains.



(a) The wavelet spatial encoding for x and y wavevectors.



(b) The FFT of the above wavelet spatial encodings.

Figure 5: The wavelet encoding of the x and y wavevectors (Fig. 5a) and its FFT (Fig. 5b). The physical wavevectors are shown in the subplot titles. For each (k_x, k_y) pair there is a unique mapping in both the spatial and spectral domains.

269 When encoding continuous constants with oscillatory functions on a finite grid, it
 270 is important to verify that distinct parameter values do not alias into nearly identical
 271 patches [21]. Otherwise the model cannot distinguish neighboring wavevectors, and
 272 mode selection collapses. We therefore checked pairwise cosine similarity of the
 273 mean-centered $\log_{10} |\text{FFT}|$ spectra of all 325 IBZ wavevector encodings produced by
 274 Algorithm 2 (Supplementary Information). The resulting similarity matrix is shown
 275 in Figure 6; the maximum off-diagonal similarity is 0.862, well below near-duplicate
 276 levels, confirming that the chosen encoding constants introduce no appreciable aliasing
 277 across the IBZ grid.

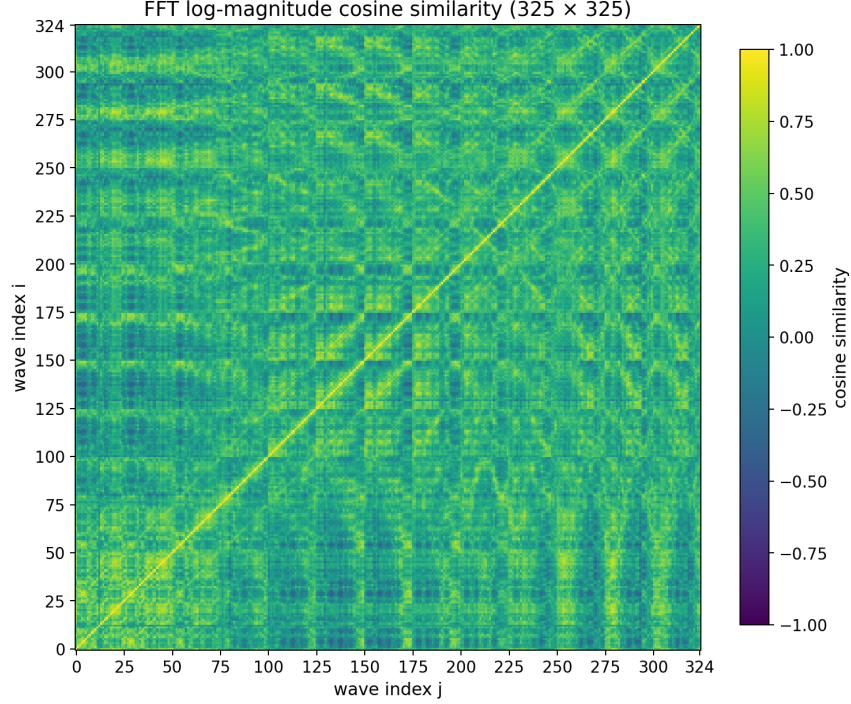


Figure 6: Pairwise cosine similarity of mean-centered $\log_{10} |\text{FFT}|$ spectra for the 325 IBZ wavevector encodings generated by Algorithm 2. The maximum off-diagonal similarity is 0.862, indicating that distinct (k_x, k_y) pairs remain spectrally separable on the 32×32 grid.

2.5. Dataset Construction

Our full dataset contains 24,000 geometries, randomly generated as described in prior subsections. Half are discrete-valued, with only 0s or 1s in each pixel location representing one of the two materials, while the other half are continuous-valued, with any value in $[0, 1]$ at each pixel representing a blended material. Each geometry has six bands and 325 wavevectors per band (a mesh grid of the IBZ half-plane), resulting in $24000 \times 6 \times 325 = 46,800,000$ sample pairs of inputs with shape $(3 \times 32 \times 32)$ and outputs of shape $(5 \times 32 \times 32)$. The data are stored as float16 PyTorch tensors.

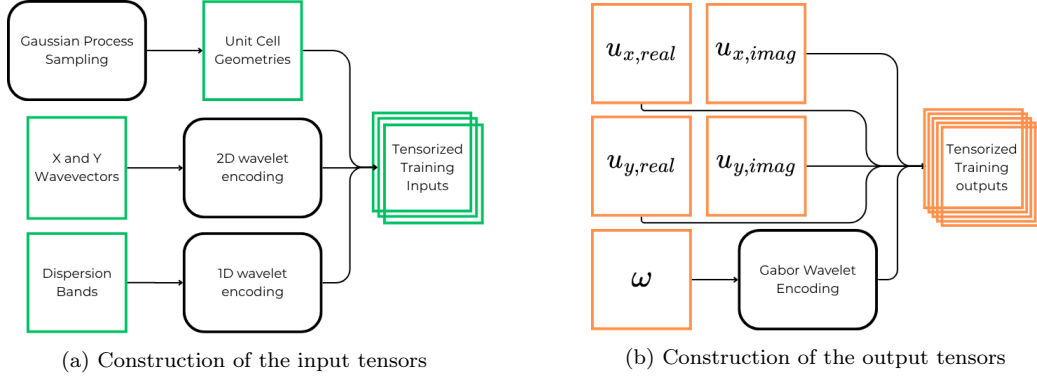


Figure 7: Subfigures a and b show the processes by which input data from Section 2.2 and output data from Section 2.3 are tensorized for ingestion by the FNO model. Note the wavelet embedding steps applied to fundamentally scalar quantities.

2.6. Training Procedure

The Fourier Neural Operator was trained using supervised learning on paired inputs and outputs described in Section 2. Each training sample consists of a $3 \times 32 \times 32$ input tensor (geometry and wavelet embeddings) and a $5 \times 32 \times 32$ output tensor (displacement field components and eigenfrequency).

Six loss functions were evaluated as training objectives: mean absolute error (MAE), mean squared error (MSE), normalized MAE, normalized MSE, Huber (smooth L1), and structural similarity index (SSIM). Losses were aggregated across all five output channels. Huber loss yielded the best overall performance and was used for all reported results; its definition is given below. Details on the other loss functions are provided in the Supplementary Information.

$$\mathcal{L}_{\text{Huber}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 \begin{cases} \frac{1}{2} (\hat{u}_c(i, j) - u_c(i, j))^2, & |\hat{u}_c(i, j) - u_c(i, j)| \leq \delta, \\ \delta \left(|\hat{u}_c(i, j) - u_c(i, j)| - \frac{\delta}{2} \right), & |\hat{u}_c(i, j) - u_c(i, j)| > \delta. \end{cases}$$

where $H = W = 32$ (pixels) and $c \in \{0, 1, 2, 3, 4\}$ corresponds to the 5 output channels: eigenfrequency, followed by real and imaginary components of x and y displacements.

For training the FNO model, combinations of model layers, hidden channels, activation function, loss function, optimizer, and training hyperparameters were evaluated. The AdamW optimizer [22] achieved the best performance compared with Adam [23], other momentum-based optimizers, and SGD. The candidate settings and selected values are reported in Table S1 in Section S1, with the selected values italicized.

The selected architecture uses four Fourier layers, 128 hidden channels, and GELU activations. Increasing the depth or width beyond these values did not appreciably improve validation performance and increased both training cost and the tendency to overfit. The selected training configuration uses Huber loss throughout, AdamW optimization, a learning rate of 2×10^{-3} , a batch size of 512, and 12 epochs. All results reported in this article were generated using this configuration.

Final model performance did not appreciably change with modest variations in scheduler step size or batch size, so these parameters were fixed at 4 and 512, respectively, for the remaining training runs. Learning rate and time to convergence were inversely related: higher learning rates accelerated convergence but produced less stable optimization. In particular, learning rates above approximately 10^{-2} occasionally caused an abrupt, irreversible increase in loss, after which training plateaued. We therefore selected a learning rate of 2×10^{-3} , which provided the best balance between convergence speed and stability.

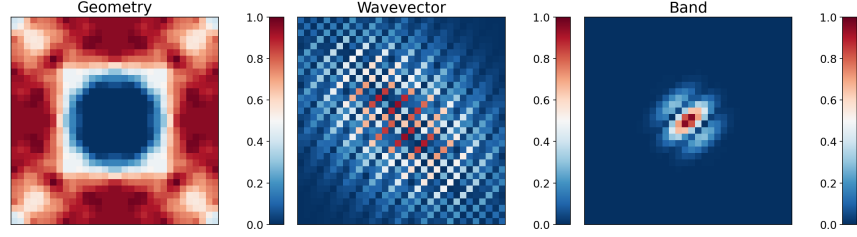
3. Results and Discussion

3.1. Displacement Field Prediction

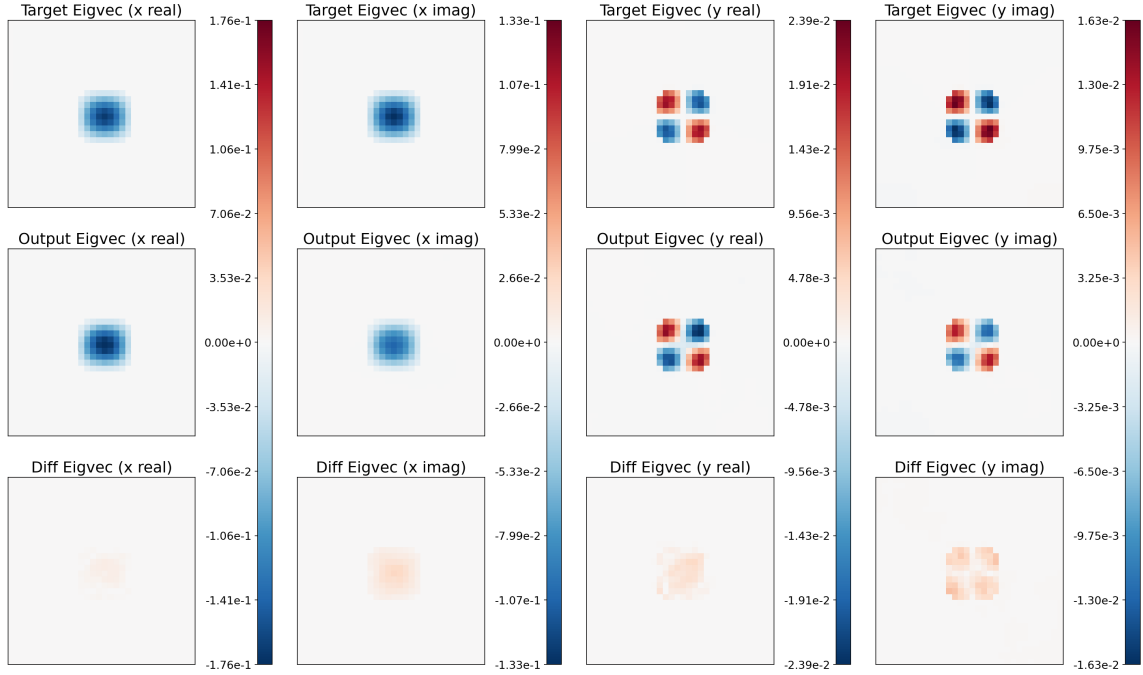
We next evaluate the trained Fourier Neural Operator on its predictions of the Bloch displacement fields (u_x, u_y) . For each unseen test sample, the prediction error is measured by the normalized mean absolute error (NMAE) over the four real-valued displacement channels, $(u_{x,\text{real}}, u_{x,\text{imag}}, u_{y,\text{real}}, u_{y,\text{imag}})$. Samples are ranked by performance, defined inversely with NMAE, so larger losses correspond to lower performance percentiles. We select representative cases at the 25th, 50th, and 75th performance percentiles, corresponding to relatively weak, median, and relatively

strong predictions. The same percentile comparison is shown for both the binarized and continuous material datasets, allowing direct visual assessment of how well the model recovers the spatial structure of the displacement modes across contrasting geometry regimes.

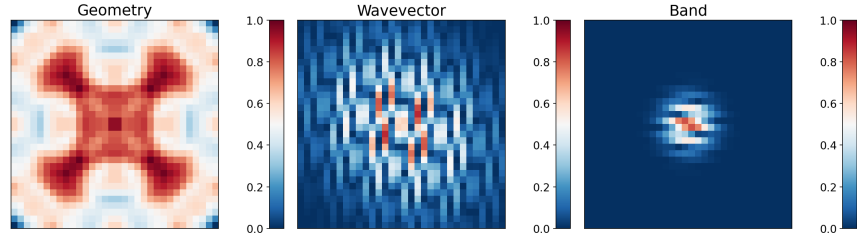
3.1.1. Continuous Geometries



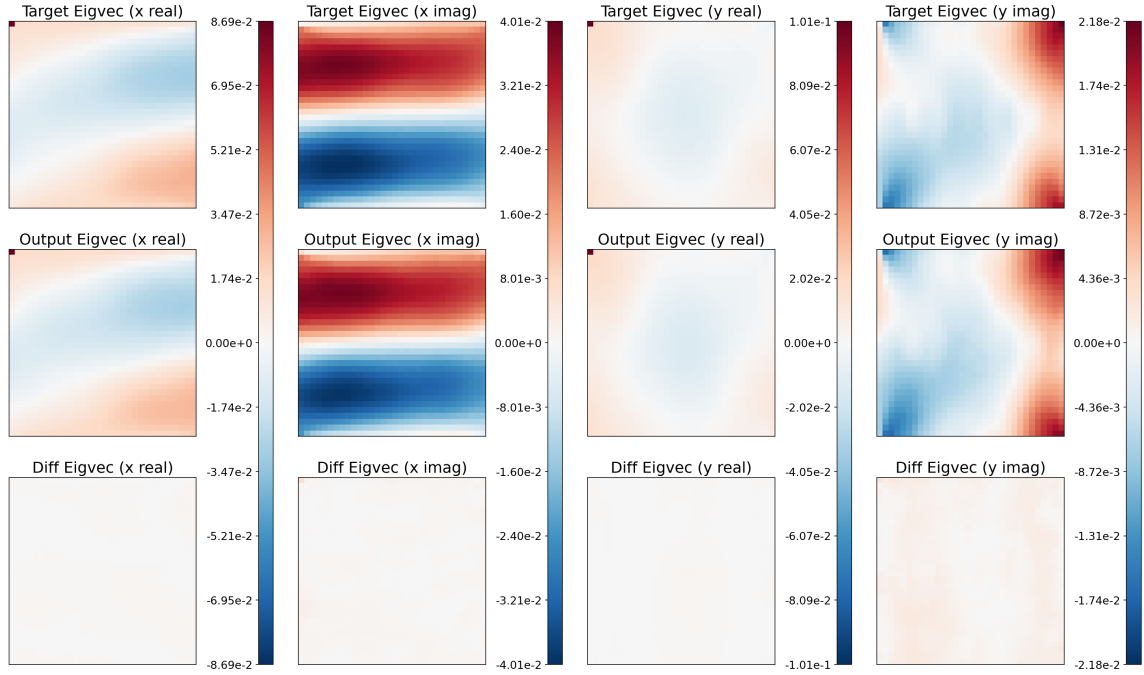
(a) Input sample at the 25th percentile of performance for unseen continuous-valued geometries.



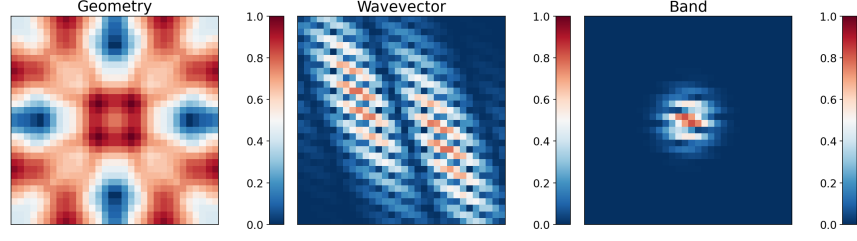
(b) Output predictions at the 25th percentile of performance for unseen continuous-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. At the 25th percentile, prediction outcomes are already fairly good, with accurate representations of scale and structure.



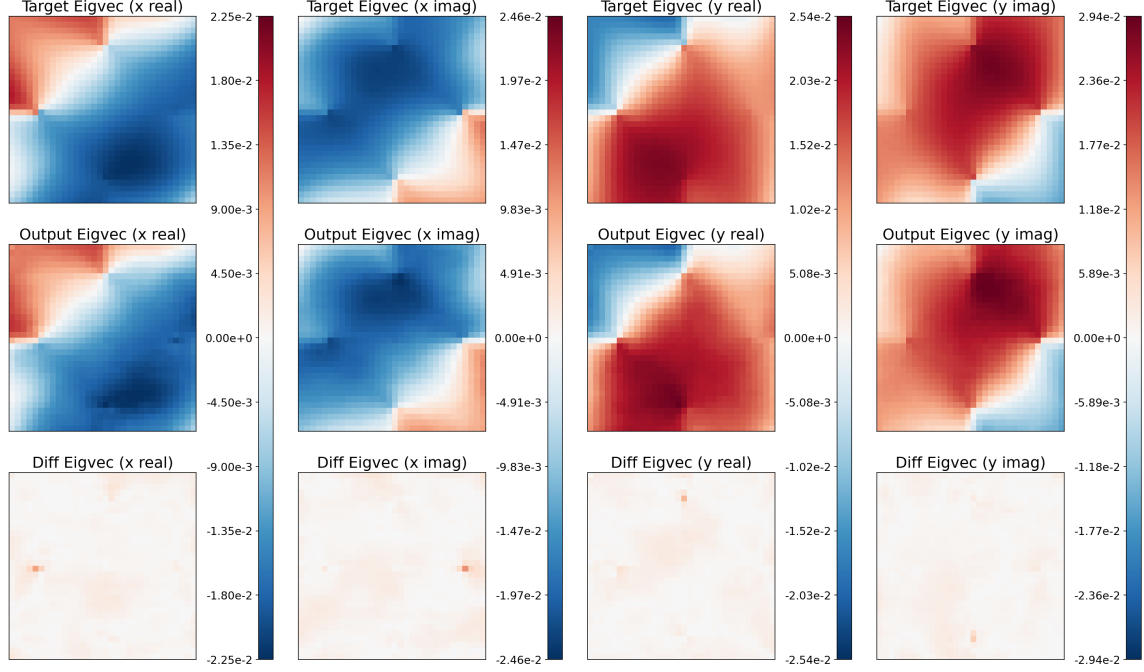
(a) Input sample at the 50th percentile of performance for unseen continuous-valued geometries.



(b) Output predictions at the 50th percentile of performance for unseen continuous-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. Predictions are able to consistently capture complex structures without graininess.

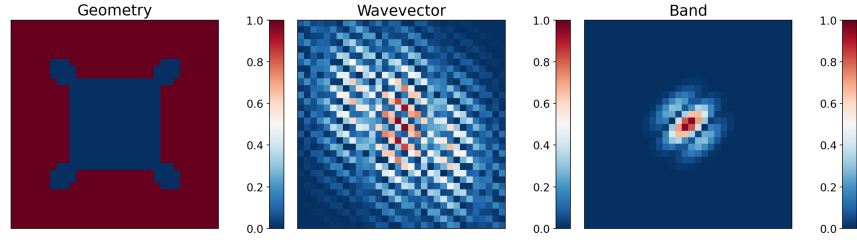


(a) Input sample at the 75th percentile of performance for unseen continuous-valued geometries.

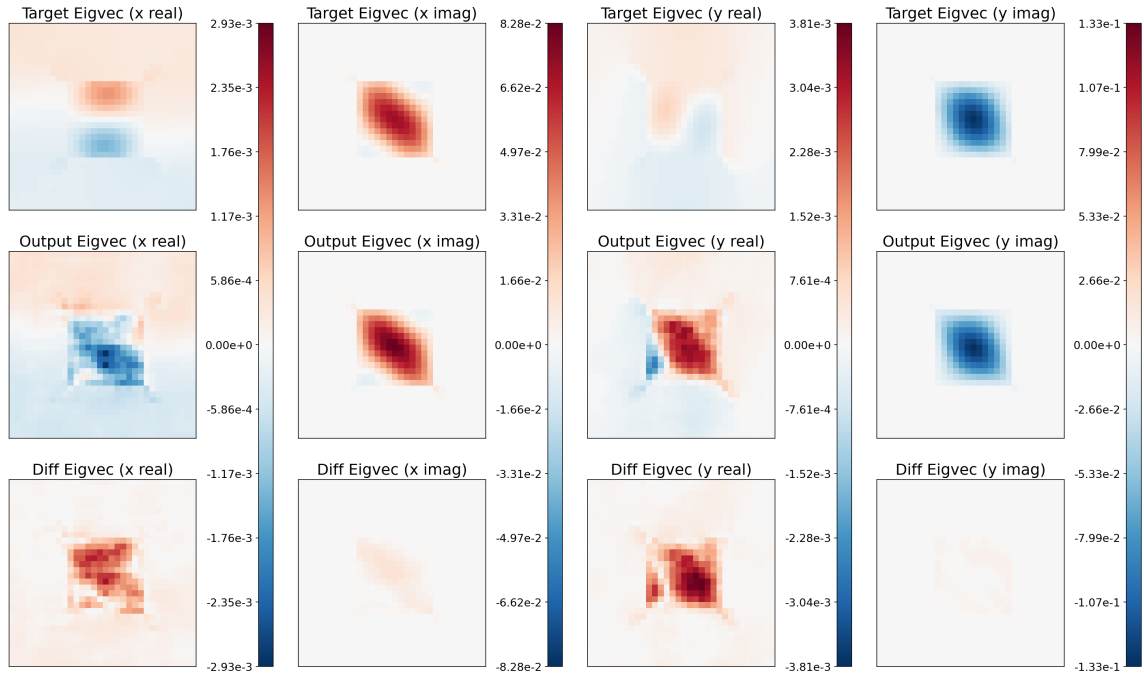


(b) Output predictions at the 75th percentile of performance for unseen continuous-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. At this point, the target and prediction are virtually indistinguishable by eye.

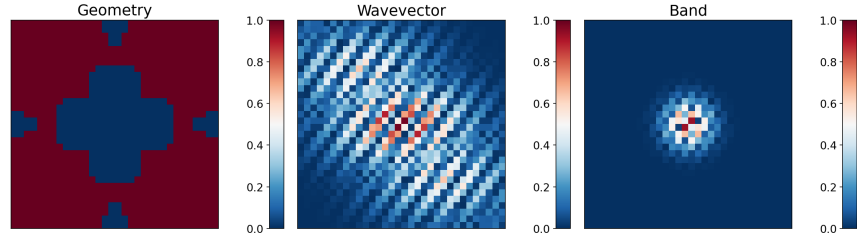
334 These results demonstrate that with wavelet encodings the FNO performs well
 335 on continuous geometries, faithfully reproducing the displacement fields in most
 336 cases. Model predictions in the upper performance quartiles match the targets well
 337 in structure and scale, with average pixel relative errors decreasing smoothly from
 338 on the order of 10^{-2} to 10^{-4} as performance percentile increases (see Figure 14a for
 339 detailed statistics).



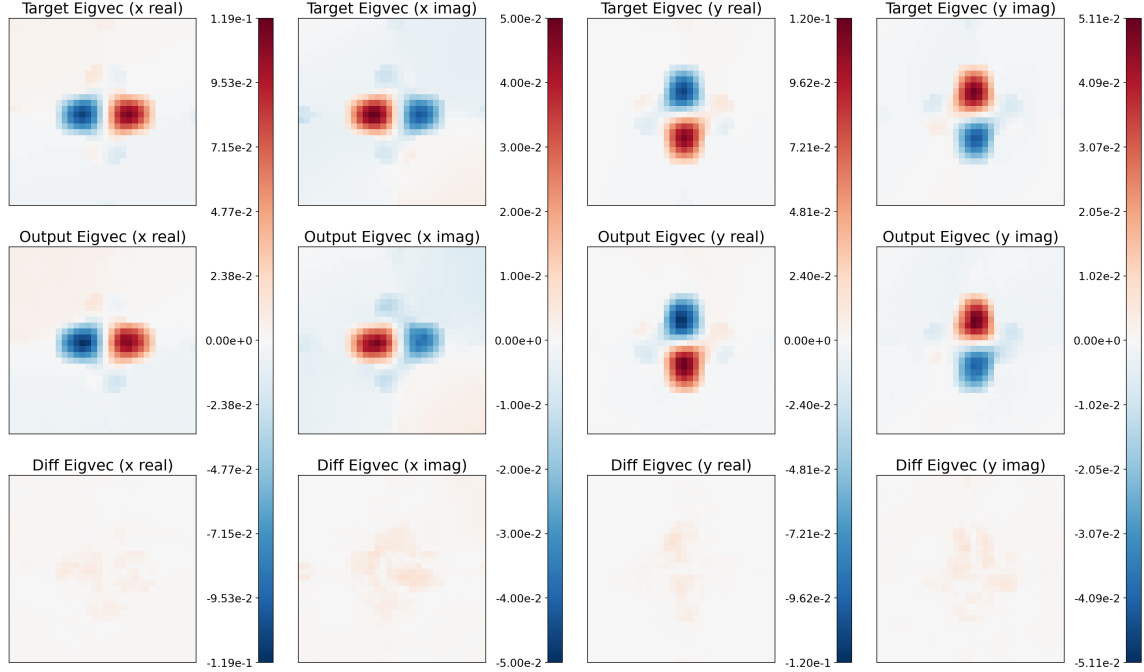
(a) Input sample at the 25th percentile of performance for unseen discrete-valued geometries.



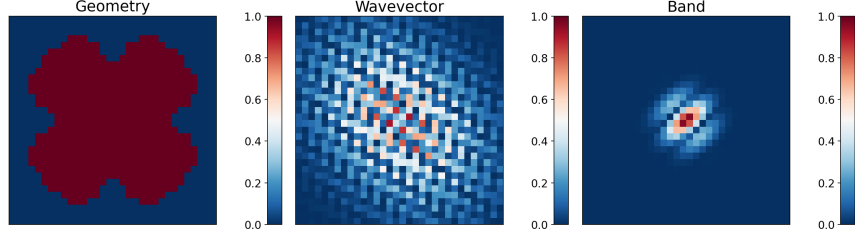
(b) Output predictions at the 25th percentile of performance for unseen discrete-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. Because Huber loss penalizes absolute residuals, for this sample the model prioritizes the two imaginary displacement channels, which are one to two orders of magnitude larger than the real channels. Absolute error is therefore low overall, but relative error remains large on the smaller-magnitude channels. (Note that the colorbars for each column are independently scaled.)



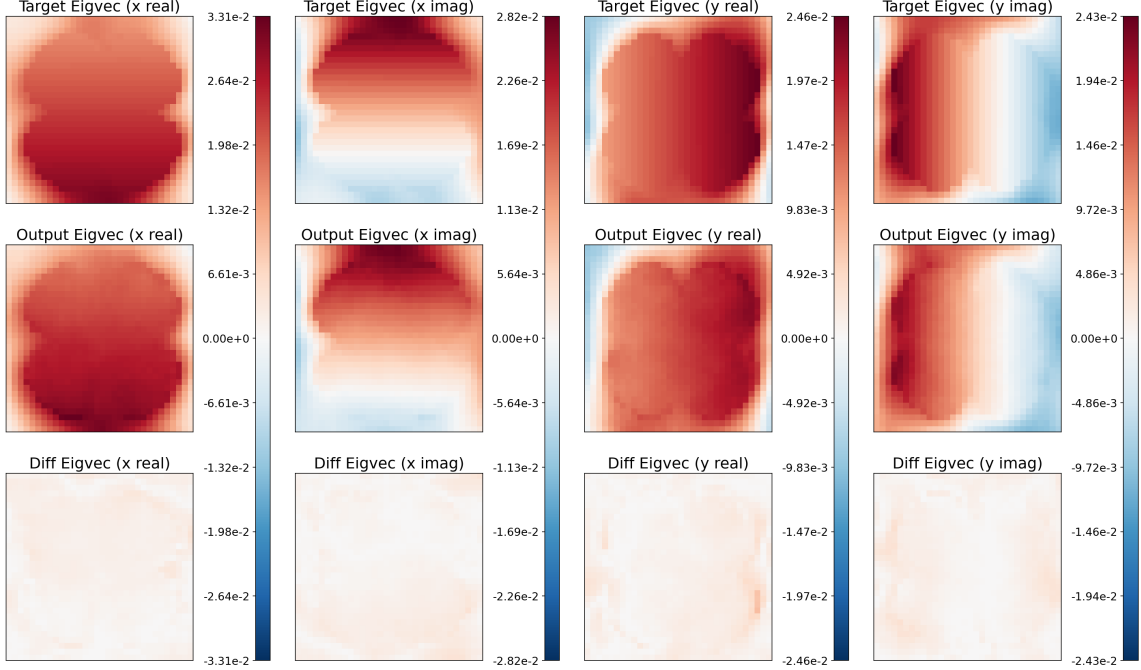
(a) Input sample at the 50th percentile of performance for unseen discrete-valued geometries.



(b) Output predictions at the 50th percentile of performance for unseen discrete-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. Absolute and relative errors are low, with strong agreement in structure and scale between targets and predictions.



(a) Input sample at the 75th percentile of performance for unseen discrete-valued geometries.



(b) Output predictions at the 75th percentile of performance for unseen discrete-valued geometries. The first row shows target fields, the second row shows predictions, and the third row shows difference fields. At this point, disagreements between targets and predictions become very hard to see by eye.

341 As in the continuous-geometry case, these results demonstrate that with wavelet
 342 encodings the FNO performs well on discrete geometries, faithfully reproducing
 343 the displacement fields in most cases. Model predictions in the upper performance
 344 quartiles match the targets well in structure and scale, with average pixel relative
 345 errors decreasing smoothly from on the order of 10^{-2} to 10^{-4} as performance percentile
 346 increases (see Figure 14b for detailed statistics).

347 3.1.3. Continuous Versus Discrete Performance

348 That a single model predicts well on both continuous and discrete geometries is a
349 nontrivial result for FNOs paired with wavelet encodings. Prior encoding strategies
350 based on constant fields and sinusoidal fields performed far worse, producing grainy
351 structures or mode collapse, because conditioning information for the wavevector and
352 band could not propagate effectively through the spectral branch of the FNO layers.
353 Performance over all unseen samples in the test set is shown in Figure 14.

354 The poorer performance on binarized geometries is consistent with basic operator-
355 learning theory and known spectral limitations of Fourier-parameterized networks [24,
356 20, 25]. Continuous material fields lie in smoother function classes, whereas binary two-
357 phase designs contain jump discontinuities with nonexistent derivatives at material
358 interfaces. Since an FNO approximates the FEA operator $\mathcal{G} : \mathcal{U} \rightarrow \mathcal{V}$ through
359 truncated Fourier representations, it is naturally advantaged when both inputs and
360 outputs are comparatively smooth. Discrete geometries thus perform worse due to
361 sharp boundaries, localized gradients, and broadband high-wavenumber content that
362 are poorly captured on a fixed 32×32 grid and are mismatched to smooth wavelet
363 encodings. Additionally, continuous geometries more often contain large regions of
364 near-zero displacement, which inflate relative error through a small denominator.

365 These results indicate two important takeaways. The first is that FNOs with
366 wavelet encodings can learn multiple deformation modes for each geometry, which
367 the user can select by setting the wavevector and band inputs. The second is that
368 the same model can learn across continuous and discrete geometry distributions,
369 suggesting that some aspects of the underlying eigenvalue problem may be captured
370 in a geometry-agnostic way. Exploring that possibility more systematically is left for
371 future work.

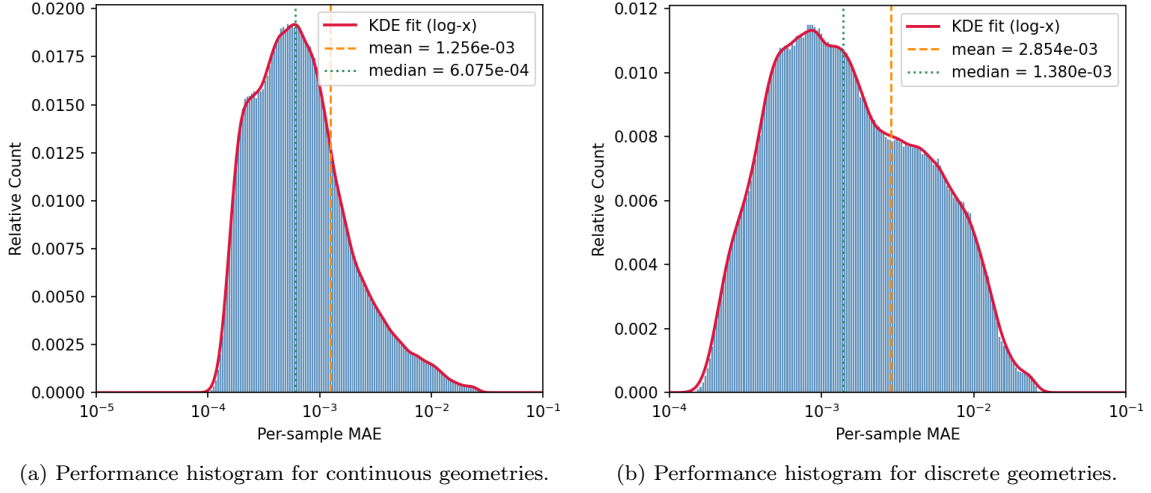


Figure 14: Comparison of relative prediction error distributions for continuous-valued and discrete metamaterial geometries for the same model. While there are some tail outliers, most samples fall within the range of 10^{-2} to 10^{-4} , indicating that the same model is able to learn continuous and discrete geometry cases.

3.2. Dispersion Band Reconstruction

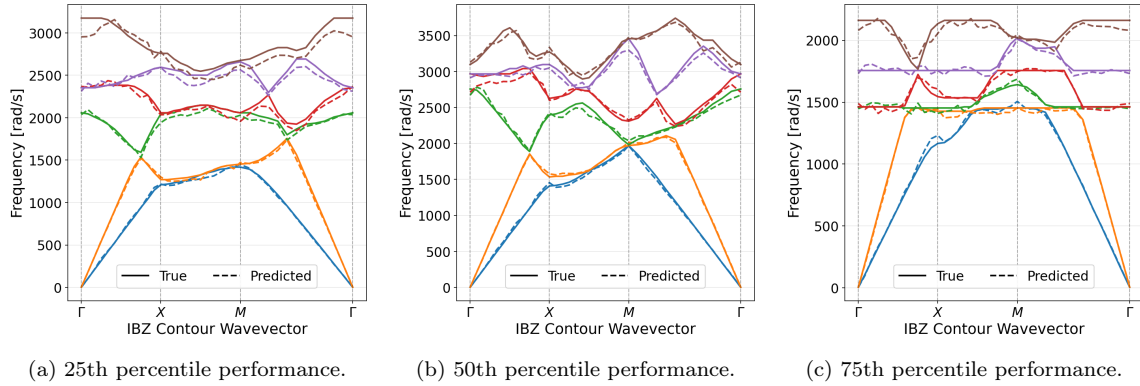


Figure 15: Predicted and ground-truth dispersion bands for representative unseen continuous-valued geometries spanning the 25th, 50th, and 75th percentiles of prediction performance.

These plots show eigenfrequency predictions across all 325 wavevectors of the IBZ half-plane for continuous-valued geometries, restricted to the horizontal, vertical, and diagonal IBZ traversals. Because the eigenfrequency is encoded with a logarithmic

transform to support multi-scale learning, errors tend to scale with the target magnitude, and higher bands can appear more deviant on a linear plot. Overall, the model achieves an average prediction error below 1% on the encoded eigenfrequency across unseen samples.

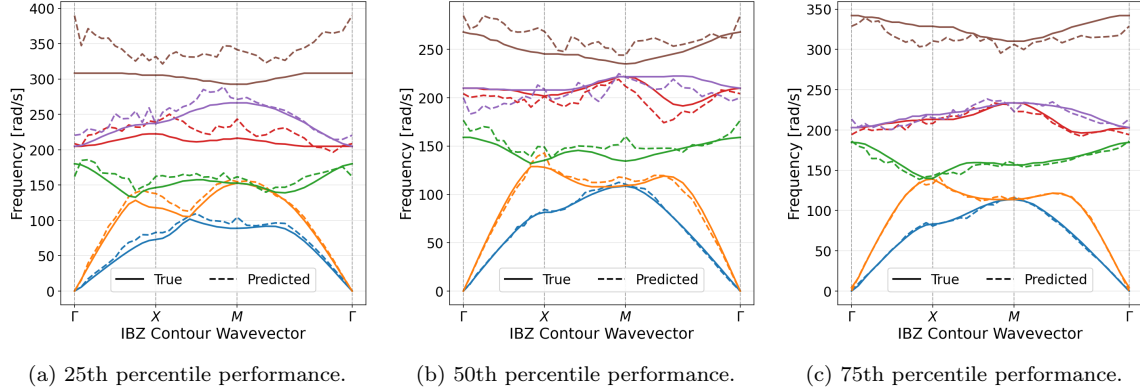


Figure 16: Predicted and ground-truth dispersion bands for representative unseen discrete-valued geometries spanning the 25th, 50th, and 75th percentiles of prediction performance.

As in the continuous-geometry figures, these plots show eigenfrequency predictions across all 325 wavevectors of the IBZ half-plane for discrete-valued geometries, restricted to the horizontal, vertical, and diagonal IBZ traversals. Overall, the model again achieves an average prediction error below 1% on the encoded eigenfrequency across unseen samples.

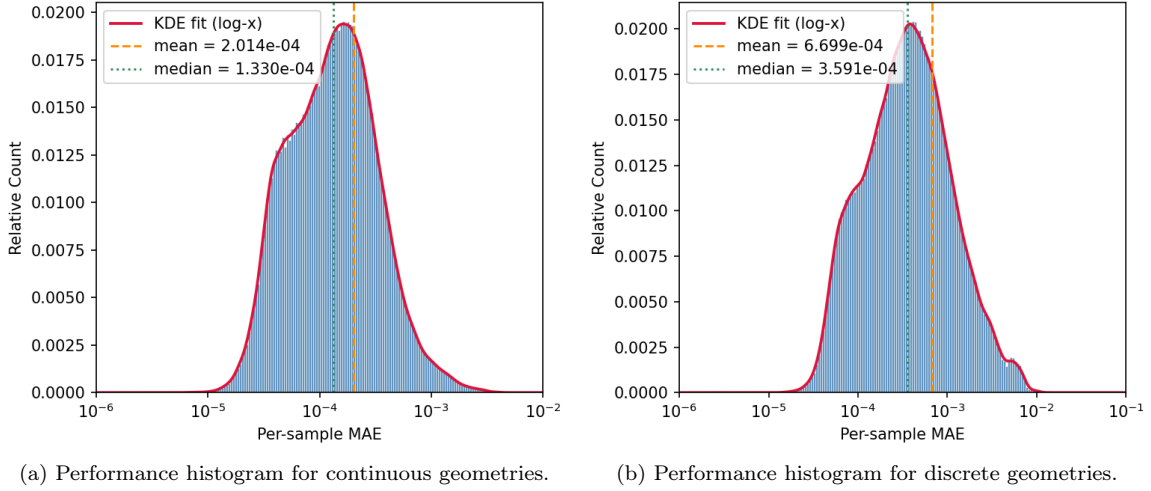


Figure 17: Comparison of relative prediction error distributions for continuous-valued and discrete metamaterial geometries for the same model. While there are some tail outliers, most samples fall within the range of 10^{-2} to 10^{-4} , indicating that the same model is able to learn continuous and discrete geometry cases.

3.3. Dependence of Prediction Error on Band and Boundary Length

The preceding subsections showed that, although the same wavelet-conditioned FNO can predict displacement fields for both continuous and discrete unit cells, discrete geometries are systematically more difficult, with absolute and relative errors larger (Figures 14, 17), and the harder quartiles of the discrete test set degrading more visibly than their continuous counterparts. This gap is expected from the architecture of the FNO itself and from the well-documented preference of spectral networks for low-frequency content [24, 20]. Each Fourier layer retains only a truncated set of spectral modes after the discrete FFT. Smooth continuous material fields, and the comparatively smooth displacement modes they induce, concentrate their energy in a modest number of low-to-moderate wavenumbers and are therefore well represented by that truncated Fourier basis. Binary geometries, by contrast, contain jump discontinuities at material interfaces. Under an FFT, such discontinuities produce a broadband spectrum whose coefficients decay slowly with wavenumber, so a substantial fraction of the high-frequency content needed to resolve sharp phase boundaries lies outside the retained modes; related analyses of Fourier-based solvers

401 for discontinuous coefficients likewise highlight Gibbs-type artifacts and degraded
 402 high-wavenumber recovery [25]. The spectral branch therefore sees a degraded, band-
 403 limited version of the interface, yielding Gibbs-type ringing or blur and a harder
 404 operator-learning problem precisely when geometric discontinuity is more severe. If
 405 this spectral-truncation explanation is correct, then even among discrete geometries
 406 the prediction error should increase with the boundary interface length. To test this
 407 empirically, we define the *boundary length* of each binarized unit cell as the number of
 408 interior four-connected pixel edges separating a 0-valued pixel from a 1-valued pixel.
 409 This scalar is a discrete measure of interface complexity. For each of the 1000 unseen
 410 discrete test geometries, we compute the mean absolute error (MAE) of the predicted
 411 displacement channels, averaged over all 325 IBZ wavevectors, separately for each of
 412 the six retained eigenbands. The resulting relationships are shown in Figure 18.

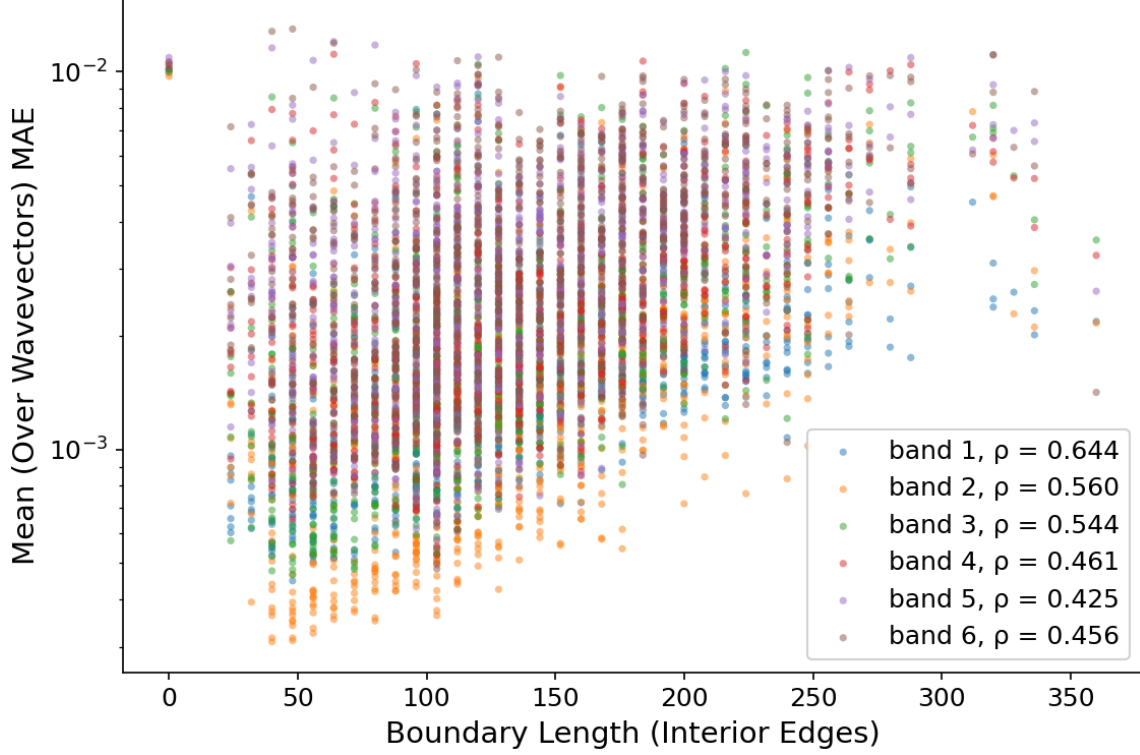


Figure 18: Mean displacement MAE versus interface boundary length, separated by color for each of the six eigenbands on the discrete test set. Points correspond to average performance on unit-cell geometries. Legend values ρ are Spearman rank correlations. A positive trend appears for every band, consistent with degraded FNO accuracy as the total length of 0/1 material interfaces increases.

Two complementary trends are apparent. First, for every band there is a positive monotonic association between boundary length and MAE, with Spearman correlations ranging from $\rho = 0.644$ for band 1 to $\rho \approx 0.42$ – 0.56 for the higher bands. Geometries with short interfaces are concentrated at lower absolute error, while geometries with long, meandering phase boundaries systematically occupy the upper portion of the cloud. This supports the interpretation that the hard cases for the learned operator are those for which the truncated Fourier representation must resolve more extensive sharp material transitions on the fixed 32×32 grid. Second, absolute error itself increases with band index. Averaged over the test geometries, the mean wavevector-averaged MAE rises from approximately 1.6×10^{-3} for band 1 to approximately 3.8×10^{-3} for band 6. Higher bands correspond to more rapidly oscillating

424 Bloch modes that concentrate energy nearer material interfaces and therefore inherit
 425 more of the geometric difficulty encoded by boundary length. Together with the
 426 continuous-versus-discrete comparisons above, Figure 18 indicates that prediction
 427 difficulty on discrete designs is organized both by band (modal complexity) and by
 428 interface measure (geometric complexity), as expected when a spectral surrogate
 429 retains only a portion of the broadband Fourier content generated by discontinuities.

430 3.4. Band and Wavevector Encoding

431 For the FNO to ingest the conditioning constants of the eigenvalue PDE in equation
 432 6, namely the wavevectors k_x , k_y and the band index b that selects which eigenmode
 433 to return, these scalars must be represented as matrices or tensors compatible with
 434 the existing channel wise input structure. Alternatives that inject scalars through
 435 specialized pathways would require heavy architectural alterations, which are hard
 436 to justify given that FNO theory suggests eigenvalue PDEs should be solvable with
 437 existing hyperparameter combinations. We therefore compared three strategies for
 438 encoding these constants as input fields: constant valued fields, spatial sinusoids, and
 439 the Gabor wavelet encoding described in Section 2.4.

440 The constant field representation is the naive default: k_x , k_y , and b are each
 441 broadcast to a constant $S \times S$ channel with $S = 32$. For the wavevectors, whose
 442 physical values lie in $[-\pi, \pi]$, these constants were normalized to the interval $[-1, 1]$
 443 before broadcasting. Band indices $\{1, 2, 3, 4, 5, 6\}$ were likewise normalized to the
 444 range $[0.1, 0.6]$. This representation fails by mode collapse, with the FNO outputting
 445 near constant fields regardless of conditioning inputs. The cause is spectral: the FFT
 446 of a constant field is zero everywhere except at the DC coefficient, so the frequency
 447 domain branch of each Fourier layer receives no information to propagate or train on
 448 past the first layer. The Fourier branch therefore cannot contribute to the prediction,
 449 leaving only the shallow spatial domain branch, which behaves like a shallow CNN
 450 and is unable to capture the complexity of the eigenvalue problem.

451 A spatial sinusoidal encoding restores unique and nondegenerate representations in
 452 the spectral domain and allows the FNO to learn the problem. On pixel coordinates

453 $(x, y) \in \{0, \dots, S - 1\}$, wavevector and band were encoded as

$$I_k(x, y) = \sin\left(\frac{2}{S}(k_x x + k_y y)\right), \quad k_x, k_y \in [-\pi, \pi].$$

454

$$I_b(x, y) = \frac{1}{2} \left[\cos\left(\frac{2\pi b x}{S}\right) + \cos\left(\frac{2\pi b y}{S}\right) \right], \quad b \in \{1, 2, 3, 4, 5, 6\}.$$

455 The wavevector field is a directed plane wave: its phase gradient is parallel to (k_x, k_y) ,
 456 so wavefronts propagate along the physical wavevector, and the spatial frequency
 457 scales with $|\mathbf{k}|$. The prefactor $2/S$ places about one cycle across the patch at the
 458 IBZ boundary $|\mathbf{k}| = \pi$, which keeps neighboring (k_x, k_y) pairs visually and spectrally
 459 distinct on the $S \times S$ grid while avoiding severe aliasing. A sine rather than a cosine is
 460 used so that the Γ point ($\mathbf{k} = \mathbf{0}$) maps to the zero field instead of a constant DC patch.
 461 However, because a sinusoid is spectrally sparse, with energy concentrated at only a
 462 few wavenumbers, the Fourier branch remains weakly utilized, and predictions exhibit
 463 a persistent grainy, under-resolved structure relative to ground truth. Increasing
 464 model depth and width did not remove this graininess, indicating that the bottleneck
 465 was the encoding rather than model capacity.

466 The Gabor wavelet encoding resolves this by design. Following Gabor’s classical
 467 construction [21], wavelets are constructed to trade certainty between the spatial
 468 and spectral domains: neither representation is infinitely sharp, but both remain
 469 informationally rich. As a result, the conditioning inputs populate both branches of
 470 each Fourier layer with usable structure, both branches can train and contribute, and
 471 the model yields the accurate, well resolved predictions reported above while enabling
 472 deterministic mode selection. These observations support the broader conclusion
 473 that input encodings should be matched to the spatial and spectral structure of the
 474 operator: encodings that populate only one domain underuse the FNO’s Fourier
 475 branch, whereas wavelet encodings exercise both.

476 4. Conclusions

477 This work demonstrates that a single Fourier Neural Operator can learn multiple
478 eigenmodes of the elastic wave equation across both continuous and discrete meta-
479 material geometries, recovering the eigenvectors as complex displacement fields and
480 the eigenvalues as eigenfrequencies. Relative to prior surrogates that primarily target
481 dispersion curves, transmission spectra, or bandgap scalars [12, 13, 14], the present
482 model returns full Bloch displacement eigenfields with deterministic multi-mode selec-
483 tion. For unseen geometries of both types, displacement predictions achieve relative
484 errors typically between 10^{-2} and 10^{-4} and dispersion reconstructions average below
485 1% error, showing that one model can span two very different geometry distributions
486 and output predictions of multiple orders of magnitude while allowing the user to
487 select any eigenmode deterministically and repeatably through arbitrary choice of
488 conditioning inputs. This capability crucially rests on how the conditioning constants
489 are encoded. Constant field encodings collapse to a single DC coefficient in the
490 spectral domain, preventing the Fourier branch of the FNO from propagating informa-
491 tion and causing mode collapse. Sinusoidal encodings are able to train meaningfully
492 but yield grainy, under-resolved predictions due to spectral sparsity. Gabor wavelet
493 encodings [21], by contrast, trade decreasing certainty in the spatial and spectral
494 representations to maintain informationally rich representations in both domains,
495 allowing both branches of each Fourier layer to richly interact with the data, leading
496 to superior learning speed and prediction performance.

497 The same spectral perspective explains the limits we observe. Prediction error
498 grows with the interface length of discrete geometries and with band index, consistent
499 with the truncated Fourier representation struggling to resolve the broadband content
500 generated by sharp material interfaces and rapidly oscillating modes [24, 20, 25].
501 Practitioners applying FNOs to interface-rich problems should therefore expect
502 accuracy to track geometric discontinuity. Even so, the trained model is lightweight,
503 roughly 2 GB at float16 precision, and evaluates in about 1 ms per sample on CPU,
504 compared with about 1 s per sample for the corresponding MATLAB FEA solve
505 on CPU. This makes the FNO a compelling surrogate for iterative design and

506 analysis in metamaterials and other fields governed by eigenvalue PDEs. Future
507 work includes generalization to other resolutions, physics-informed losses in the spirit
508 of physics-informed operator learning [6], coupling to interpretable bandgap design
509 workflows [18, 19], and additional conditioning parameters such as material constants,
510 expanding the accessible problem and solution space.

511 **AI Usage Disclosure**

512 Generative AI tools were used to assist with coding of experiments, translating
513 MATLAB code to Python, and automating training pipelines. AI tools were also used
514 for light editing of the manuscript. All scientific claims, results, and final wording
515 were reviewed and approved by the authors.

516 **Declaration of Competing Interest**

517 The authors declare that they have no known competing financial interests or
518 personal relationships that could have appeared to influence the work reported in
519 this paper.

520 **CRedit authorship contribution statement**

521 **Han Zhang:** Conceptualization, Data curation, Formal analysis, Investigation,
522 Methodology, Software, Validation, Visualization, Writing – original draft, Writing –
523 review and editing.

524 **Alexander Ogren:** Data curation, Software.

525 **Cynthia Rudin:** Supervision, Writing – review and editing.

526 **Johann Guilleminot:** Formal analysis, Methodology, Supervision, Writing – review
527 and editing.

528 **L. Catherine Brinson:** Conceptualization, Funding acquisition, Methodology,
529 Project administration, Resources, Software, Supervision, Visualization, Writing –
530 review and editing.

531 Data availability

532 Python code and experiments can be found at [https://github.com/trutheresy/](https://github.com/trutheresy/NO-2D-Metamaterials)
533 NO-2D-Metamaterials.

534 MATLAB FEA code can be found at <https://github.com/aco8ogren/2D-dispersion>.
535 git.

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617 S1. Supplementary Information

618 S1.1. Model and Training Hyperparameters

Hyperparameter	Candidate Settings			
Fourier Layers	4	6	8	12
Hidden Channels	32	64	128	256
Activation Function	ReLU	GELU	tanh	Sigmoid
Loss Criterion	L1	L2	Huber	SSIM
Epochs	8	12	16	20
Learning Rate	10^{-4}	2×10^{-3}	10^{-2}	10^{-1}
Weight Decay	0	10^{-4}	10^{-3}	10^{-2}
Gamma	0	0.01	0.1	0.5
Step Size	0	1	4	10
Batch Size	64	128	256	512
Data Precision	F8 E4M3	F8 E5M2	F16	F32

Table S1: Combinations tried in grid search for optimal model and training hyperparameters.

619 S1.2. Loss Criterion

620 In the exploration of optimal training protocols for the FNO model, several
621 loss functions were evaluated as training objectives: mean absolute error (MAE),
622 mean squared error (MSE), normalized MAE, normalized MSE, Huber (smooth L1),
623 and structural similarity index (SSIM). For comparison, validation performance was
624 measured with MAE and MSE. Broadly speaking, models that performed best in MSE
625 also performed best in MAE. The training objective that yielded the lowest MAE and
626 MSE was Huber loss, which was used for all reported results. All comparisons used
627 a learning rate of 2×10^{-3} with a per-epoch decay factor of 0.9. The mathematical
628 definitions of each loss follow. Losses were evaluated and aggregated across all five
629 output channels.

- 630 • The mean absolute error (MAE) loss is defined as

$$\mathcal{L}_{\text{MAE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 |\hat{u}_c(i, j) - u_c(i, j)|$$

Experimentally, models trained exclusively with MAE produced predictions with sharper interfaces and better-preserved boundary features than those trained solely with MSE. This behavior is consistent with the linear penalty of MAE, which does not disproportionately penalize isolated large residuals near discontinuities. Despite the improved qualitative sharpness, MAE training converged to higher validation MAE and MSE than Huber loss, indicating that the improved preservation of fine structural features came at the expense of overall numerical accuracy. This observation also motivated the hybrid training schedule in which optimization began with MAE before transitioning to MSE, combining the edge-preserving characteristics of MAE with the stronger refinement of large residuals provided by MSE.

- The mean squared error (MSE) loss is defined as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 (\hat{u}_c(i, j) - u_c(i, j))^2$$

The quadratic penalty of MSE assigns increasingly large gradients to larger residuals, causing the optimizer to preferentially reduce regions with the greatest numerical error. As a consequence, the loss favors smooth predictions that minimize the overall energy of the residual rather than preserving sharp transitions. Experimentally, models trained exclusively with MSE produced outputs that reproduced the correct magnitude and large-scale structure of the solution, but exhibited diffuse interfaces and blurred boundaries. Fine-scale features were often replaced by low-amplitude, grainy artifacts, consistent with the tendency of MSE to average over uncertain or discontinuous regions in order to minimize the global squared error.

- A hybrid MAE–MSE training schedule was also evaluated, where the model was optimized using MAE for the first half of the training epochs before switching to MSE for the remaining half. This training strategy approximates the behavior of the Huber loss over the course of optimization. During the early stages of

training, prediction errors are typically large, making the linear penalty of MAE advantageous for preserving sharp interfaces while preventing large residuals from dominating the optimization. As training progresses and the average residual decreases, switching to MSE increases the emphasis on refining the remaining prediction errors and improving overall numerical accuracy.

Experimentally, this training schedule consistently outperformed training with either MAE or MSE alone for the same total number of epochs. The resulting predictions retained substantially sharper boundaries than models trained exclusively with MSE, while achieving lower validation MAE and MSE than models trained exclusively with MAE. Among the non-Huber training strategies investigated, the hybrid schedule produced the best overall balance between qualitative feature preservation and quantitative accuracy, converging to within approximately 10% of the validation error achieved by Huber loss. Although less adaptive than Huber, which transitions between MAE- and MSE-like behavior on a per-residual basis, the epoch-wise transition captures much of the same optimization benefit.

- The structural similarity index measure (SSIM) is defined as

$$\mathcal{L}_{\text{SSIM}} = 1 - \frac{1}{5} \sum_{c=0}^4 \frac{(2\mu_{u_c}\mu_{\hat{u}_c} + C_1)(2\sigma_{u_c\hat{u}_c} + C_2)}{(\mu_{u_c}^2 + \mu_{\hat{u}_c}^2 + C_1)(\sigma_{u_c}^2 + \sigma_{\hat{u}_c}^2 + C_2)}, \quad \text{where}$$

$$\mu_{u_c} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} u_c(i, j), \quad \sigma_{u_c\hat{u}_c} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} (u_c(i, j) - \mu_{u_c})(\hat{u}_c(i, j) - \mu_{\hat{u}_c})$$

A significant limitation of SSIM arises when it is applied to signed-valued fields. Unlike its original application to nonnegative image intensities, a perfect sign-reversed field ($\hat{u} = -u$) can achieve nearly the same SSIM score as a perfect reconstruction ($\hat{u} = u$). Consequently, a model trained solely with SSIM may converge to physically incorrect solutions containing polarity inversions while still minimizing the loss. The following derivation demonstrates the mathematical origin of this ambiguity.

682 For the purposes of the following discussion, we consider the common case
 683 where the stabilizing constants satisfy

$$C_1 \ll 2\mu_{u_c}^2, \quad C_2 \ll 2\sigma_{u_c}^2,$$

684 allowing them to be neglected.

685 For a correct reconstruction,

$$u_c^{(+)} = u_c,$$

686 the SSIM score is

$$\text{SSIM}(u_c, u_c^{(+)}) = \frac{(2\mu_{u_c}^2)(2\sigma_{u_c}^2)}{(2\mu_{u_c}^2)(2\sigma_{u_c}^2)} = 1.$$

687 Now consider a sign-reversed reconstruction. The field and its mean reverse
 688 sign,

$$u_c^{(-)} = -u_c, \quad \mu_{u_c^{(-)}} = -\mu_{u_c},$$

689 while the covariance reverses sign and the variance remains unchanged,

$$\sigma_{u_c, u_c^{(-)}} = -\sigma_{u_c, u_c}, \quad \sigma_{u_c^{(-)}}^2 = \sigma_{u_c}^2.$$

690 Substituting these relationships into the SSIM expression gives

$$\text{SSIM}(u_c, u_c^{(-)}) = \frac{(-2\mu_{u_c}^2)(-2\sigma_{u_c}^2)}{(2\mu_{u_c}^2)(2\sigma_{u_c}^2)} = 1.$$

691 The numerator therefore differs from that of a correct reconstruction by two
 692 factors of -1 , and thus, under these conditions, SSIM assigns essentially the
 693 same similarity score to a field and its sign-reversed counterpart as it does to
 694 a perfect reconstruction. This ambiguity constitutes a significant limitation
 695 when SSIM is applied to signed-valued regression problems where the polarity
 696 of the solution is physically meaningful. During training, the network frequently
 697 converged to predictions in which localized regions or the entire image were

the negative of the correct solution while still achieving a favorable SSIM objective. Although the resulting fields exhibited accurate interface locations, boundary geometry, and overall morphology, the polarity inversions rendered the predictions physically incorrect. Consequently, SSIM should not be used as the sole training criterion for signed-valued fields unless it is supplemented by an additional loss that explicitly penalizes sign reversals.

- The normalized mean absolute error (NMAE) loss is defined as

$$\mathcal{L}_{\text{NMAE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 \frac{|\hat{u}_c(i, j) - u_c(i, j)|}{|u_c(i, j)| + \varepsilon}$$

Unlike MAE, NMAE weights each prediction error by the inverse magnitude of the target value, causing the optimization to place greater emphasis on regions where the ground-truth solution is small in scale. Consequently, the loss seeks to minimize relative error rather than absolute error, preventing low-magnitude features from being dominated by high-magnitude regions during training.

Experimentally, NMAE successfully reduced the relative error in low-amplitude portions of the solution. However, this improvement came at the expense of increased overall MAE and MSE compared with training using the standard MAE loss. For the present application, the absolute magnitude of the prediction error was determined to be of greater practical importance than the relative error. As a result, the improved accuracy in low-valued regions did not outweigh the degradation in overall numerical accuracy, making NMAE a less suitable optimization criterion for the objectives of this work. In other applications, however, NMAE can be a valid and sometimes preferable alternative to standard MAE.

- The normalized mean squared error (NMSE) loss is defined as

$$\mathcal{L}_{\text{NMSE}} = \frac{1}{HW} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \sum_{c=0}^4 \frac{(\hat{u}_c(i, j) - u_c(i, j))^2}{(u_c(i, j))^2 + \varepsilon}$$

721 Similar to NMAE, NMSE normalizes the prediction error by the magnitude
722 of the target field, causing the optimization to prioritize relative error rather
723 than absolute error. The quadratic penalty retains the characteristic behavior
724 of MSE by assigning greater weight to larger residuals, while the normalization
725 emphasizes regions where the target magnitude is small.

726 Experimentally, NMSE reduced the relative error in low-amplitude regions
727 compared with standard MSE. However, this improvement was accompanied by
728 increased overall validation MAE and MSE, performing worse than conventional
729 MSE optimization. As with NMAE, the objective of this work was to minimize
730 the absolute prediction error rather than the relative error. Consequently, the
731 increased emphasis placed on low-magnitude regions did not translate into
732 improved overall predictive performance, making NMSE less appropriate for
733 the present application.

734 *S1.3. Algorithms for Input Wavelet Encoding*

735 The band and wavevector conditioning channels used by the FNO are generated
736 by the Gabor wavelet embeddings below, which match the implementations in the
737 accompanying code.

Algorithm 1 1D Gabor Wavelet Embedding from Band Index

- 1: **INPUT:** Band index b ; grid size $S \leftarrow 32$; frequency scale $r \leftarrow 2.0$
- 2: Create linspace vectors $x, y \in [-1, 1]$ with S points
- 3: Construct meshgrid $X, Y \leftarrow \text{meshgrid}(x, y)$
- 4: Compute base frequency:

$$f \leftarrow (1.0 + |b|) \cdot \frac{S}{8}$$

- 5: Compute rotation angle:

$$\theta \leftarrow (b \bmod 8) \cdot \frac{S}{16}$$

- 6: Rotate coordinates:

$$X_\theta \leftarrow X \cos \theta + Y \sin \theta, \quad Y_\theta \leftarrow -X \sin \theta + Y \cos \theta$$

- 7: Set Gaussian envelope widths:

$$\sigma_x \leftarrow \sigma_y \leftarrow \frac{0.3}{r}$$

- 8: Compute Gabor wavelet embedding:

$$\psi_b(x, y) \leftarrow \exp \left(-\frac{X_\theta^2}{2\sigma_x^2} - \frac{Y_\theta^2}{2\sigma_y^2} \right) \cdot \cos(f \cdot X_\theta)$$

- 9: **RETURN:** $\psi_b \in \mathbb{R}^{S \times S}$
-

738 In Algorithm 1, the grid size $S = 32$ matches the geometry resolution used by
739 the FNO. For this resolution the base frequency is $f = (1 + |b|) S/8$, so successive
740 bands increase both carrier frequency and orientation. The modulo in θ repeats
741 orientation every eight bands, while f continues to increase, so embeddings remain
742 distinguishable beyond eight bands. The constants $r = 2.0$, $S/16$, and 0.3 were chosen
743 so that the wavelets occupy most of the spatial domain and remain visually distinct
744 across bands.

Algorithm 2 2D Gabor Wavelet Embedding for Wavevectors

- 1: **INPUT:** Wavevector components k_x, k_y (radians); grid size $S \leftarrow 32$; frequency scale $r \leftarrow 1.0$
- 2: Set constants $f_0 \leftarrow 2.20$, $\alpha \leftarrow 41$, $N_x \leftarrow 13$, $N_y \leftarrow 25$
- 3: Create linspace vectors $x, y \in [-1, 1]$ with S points
- 4: Construct meshgrid $X, Y \leftarrow \text{meshgrid}(x, y)$
- 5: Compute frequencies:

$$f_x \leftarrow (f_0 + k_x) \alpha, \quad f_y \leftarrow (f_0 + k_y) \alpha$$

- 6: Compute rotation angles:

$$\theta_x \leftarrow (k_x \bmod N_x) \cdot \frac{\pi}{N_x}, \quad \theta_y \leftarrow (k_y \bmod N_y) \cdot \frac{\pi}{N_y}$$

- 7: Rotate coordinates:

$$X_\theta \leftarrow X \cos \theta_x + Y \sin \theta_y, \quad Y_\theta \leftarrow -X \sin \theta_x + Y \cos \theta_y$$

- 8: Set Gaussian envelope widths:

$$\sigma_x \leftarrow \sigma_y \leftarrow \frac{0.5}{r}$$

- 9: Compute Gabor wavelet embedding:

$$\psi_{k_x, k_y}(x, y) \leftarrow \exp\left(-\frac{X_\theta^2}{2\sigma_x^2} - \frac{Y_\theta^2}{2\sigma_y^2}\right) \cdot \sin(f_x X_\theta) \cdot \sin(f_y Y_\theta)$$

- 10: **RETURN:** $\psi_{k_x, k_y} \in \mathbb{R}^{S \times S}$
-

745 In Algorithm 2, k_x and k_y enter as continuous radian valued wavevector com-
746 ponents. Distinct modulo periods $N_x = 13$ and $N_y = 25$ reduce aliasing under
747 interchange of the two components. The offset f_0 and scale α map the physical
748 range of k into carrier frequencies that remain well resolved on the $S \times S$ grid, while

749 $\sigma = 0.5/r$ sets the envelope width so that the patterns remain spatially localized yet
750 informative in both domains.

751 *S1.4. Wavelet Decoding*

752 For purposes of checking general applicability, an experiment was run to check
753 if a positive scalar may be embedded in a Gabor wavelet image and recovered
754 after pixelization, storage, or inference. Algorithms 3 and 4 implement this encode–
755 decode pair by spectral peak detection with centroid refinement. In the experiment,
756 we test recovery over $[1, 8000]$, spanning about four orders of magnitude. Despite
757 discretization onto a finite $S \times S$ grid and limited numeric precision associated with
758 float16 storage and arithmetic, the decoded values remain accurate across the full
759 range. Figures S1 and S2 show representative encodings and the corresponding round
760 trip error.

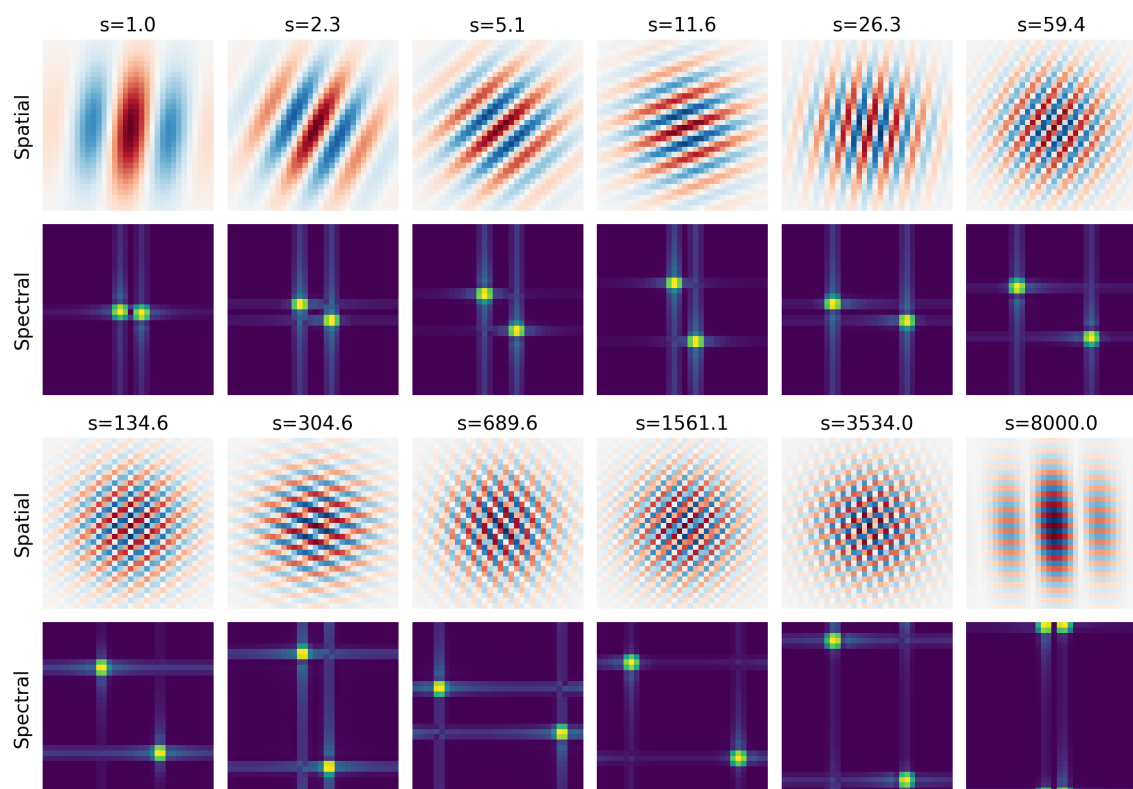


Figure S1: Log spaced Gabor wavelet encodings of scalars from 1 to 8000. Alternating rows show the spatial wavelet images and the corresponding Fourier magnitude spectra.

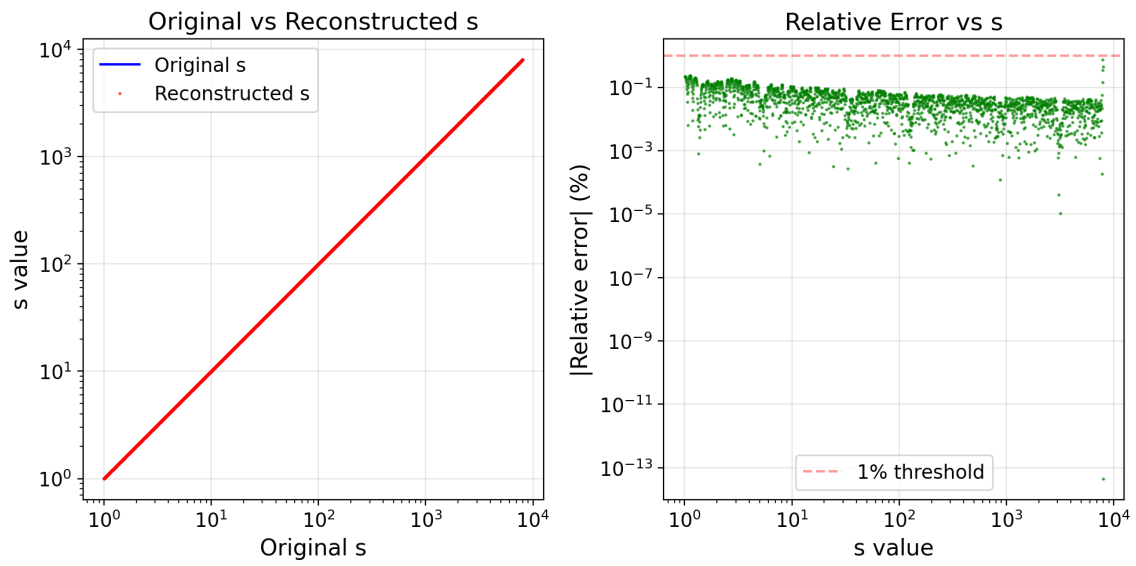


Figure S2: Encode–decode error for scalars from a minimum of 1 to a maximum of 8000.

Algorithm 3 Scalar Wavelet Encoding

- 1: **INPUT:** Scalar $s > 0$; image size $S \leftarrow 32$; bounds $[s_{\min}, s_{\max}]$; spectral-index bounds $[k_{\min}, k_{\max}]$
- 2: Set constants $\gamma \leftarrow 1$, $\phi \leftarrow 0$, $\sigma \leftarrow 8$, $\theta_{\min} \leftarrow \pi/30$, $\theta_{\max} \leftarrow \pi/2 - \pi/30$
- 3: Compute log-domain values:

$$\ell_s \leftarrow \log s, \quad \ell_{\min} \leftarrow \log s_{\min}, \quad \ell_{\max} \leftarrow \log s_{\max}$$

- 4: Compute normalized position and spectral index:

$$t \leftarrow \text{clip}\left(\frac{\ell_s - \ell_{\min}}{\ell_{\max} - \ell_{\min}}, 0, 1\right), \quad k \leftarrow k_{\min} + t(k_{\max} - k_{\min})$$

- 5: Compute log width per unit k :

$$\Delta_\ell \leftarrow \frac{\ell_{\max} - \ell_{\min}}{k_{\max} - k_{\min}}$$

- 6: Compute orientation from within-band log position:

$$\theta \leftarrow \theta_{\min} + \left(\frac{(\ell_s - \ell_{\min}) \bmod \Delta_\ell}{\Delta_\ell}\right) (\theta_{\max} - \theta_{\min})$$

- 7: Build centered grid $X, Y \in \{-S/2, \dots, S/2 - 1\}$, then rotate:

$$X_\theta \leftarrow X \cos \theta + Y \sin \theta, \quad Y_\theta \leftarrow -X \sin \theta + Y \cos \theta$$

- 8: Set envelope widths $\sigma_x \leftarrow \sigma$, $\sigma_y \leftarrow \gamma \sigma_x$, and carrier frequency $\omega \leftarrow 2\pi k/S$
- 9: Compute wavelet image:

$$\psi_s \leftarrow \exp\left(-\frac{1}{2} \left(\frac{X_\theta^2}{\sigma_x^2} + \frac{Y_\theta^2}{\sigma_y^2}\right)\right) \cos(\omega X_\theta + \phi)$$

- 10: **RETURN:** $\psi_s \in \mathbb{R}^{S \times S}$, k , θ
-

Algorithm 4 Scalar Wavelet Decoding

- 1: **INPUT:** Wavelet image $\psi \in \mathbb{R}^{S \times S}$; bounds $[s_{\min}, s_{\max}]$, $[k_{\min}, k_{\max}]$, $[\theta_{\min}, \theta_{\max}]$
- 2: Set constants for centroid refinement: radius $R \leftarrow 3$, power $p \leftarrow 2$
- 3: Mean-center image $\psi_c \leftarrow \psi - \text{mean}(\psi)$, then compute the centered 2D discrete Fourier magnitude:

$$\hat{\psi}(k_x, k_y) \leftarrow \sum_{m=0}^{S-1} \sum_{n=0}^{S-1} \psi_c(m, n) \exp\left(-2\pi i \left(\frac{k_x m}{S} + \frac{k_y n}{S}\right)\right), \quad F(k_x, k_y) \leftarrow |\hat{\psi}(k_x, k_y)|$$

- 4: Keep upper-half-plane spectral support (plus positive-frequency center row) and find dominant peak index (r^*, c^*)
- 5: Define symmetric peak partner about center, then compute weighted local centroid around (r^*, c^*) within radius R , using weights $w = F^p$ and excluding points closer to the symmetric partner
- 6: Convert refined centroid to wavevector coordinates (k_x, k_y) , then:

$$k_{\text{ext}} \leftarrow \sqrt{k_x^2 + k_y^2}, \quad \theta_{\text{ext}} \leftarrow \text{atan2}(k_y, k_x) \bmod \pi$$

- 7: Map extracted k to approximate log value:

$$\ell_{\text{approx}} \leftarrow \ell_{\min} + \text{clip}\left(\frac{k_{\text{ext}} - k_{\min}}{k_{\max} - k_{\min}}, 0, 1\right) (\ell_{\max} - \ell_{\min})$$

- 8: Let $\Delta_\ell \leftarrow (\ell_{\max} - \ell_{\min}) / (k_{\max} - k_{\min})$, compute within-band position from angle:

$$\alpha \leftarrow \text{clip}\left(\frac{\theta_{\text{ext}} - \theta_{\min}}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \quad \delta_\ell \leftarrow \alpha \Delta_\ell$$

- 9: Find nearest consistent log value by testing neighboring bands $b-1, b, b+1$:

$$\ell^* \leftarrow \arg \min_{\ell \in \{\ell_{\min} + j\Delta_\ell + \delta_\ell\}} |\ell - \ell_{\text{approx}}|$$

- 10: Clip ℓ^* to $[\ell_{\min}, \ell_{\max}]$, then decode:

$$s_{\text{ext}} \leftarrow \exp(\ell^*)$$

- 11: **RETURN:** $s_{\text{ext}}, k_{\text{ext}}, \theta_{\text{ext}}$
-

761 Algorithms 3 and 4 summarize the forward and inverse mapping used in this
762 experiment. The forward mapping places the scalar on a log scale and jointly encodes
763 magnitude and orientation into a Gabor-like pattern. The inverse mapping recovers an
764 estimate by spectral peak detection with centroid refinement, followed by a consistency
765 step that combines extracted magnitude and angle to reconstruct the most likely log
766 value.